

MAHITHA SANNALA

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EDUCATION

Master of Science - Computer Science

(Sep 2022 - Dec 2023) | USA

University of California, Riverside, California

Bachelor of Technology- Computer Science and Engineering

(Aug 2018 - Jun 2022) | India

Indian Institute of Technology, Palakkad, Kerala

Selected Coursework : Artificial Intelligence, Data mining techniques, NLP Seminar Course, Information Retrieval & Web Search, Big Data Analysis , Principles of Distributed Computing, Spatial Data Analysis, Reinforcement learning

EXPERIENCE

Software Engineer - Machine Learning, We Are Here, TX (Recommendation Systems | AWS | MERN | CRM) (Jun 2023 - Present)

- Developed an ML based recommendation system, automating the initial manual, human-analyzed recommendation process through the implementation of content-based filtering. This approach utilizes multivariate data analysis to deliver tailored suggestions.
- Architected and implemented the ML model using AWS Managed Services, employing AWS Lambda and SageMaker for model development, and EC2 and S3 for deployment and storage. Incorporated a Cron job, executed via AWS CLI, to update the product library on S3 daily, ensuring the recommendations remain accurate and up-to-date.
- Constructed a REST API with AWS API Gateway to trigger AWS Lambda function, displaying recommendation results seamlessly on the web interface.
- Developed a web interface using the MERN tech stack, enabling the business to showcase results from the model and automate data exchange with the HubSpot CRM for enhanced human interaction and efficiency.
- This pilot project achieved a 10X increase in efficiency compared to the manual effort previously required, while maintaining a reliability rate of 75-80%.
- Analyzed Member's Note and Drug Data using AWS bedrock, leveraging the LLAMA 2 model along with advanced prompt tuning and engineering techniques. This work included summarizing member conversation notes, extracting drugs using BioBERT, and identifying drug manufacturers as generic, original, or affiliated with parent organizations with LLM prompt engineering.
- Currently enhancing synthetic data generation with the MIT-developed SDV library and a Reinforcement Learning with Human Feedback (RLHF) approach to refine model recommendations and simulate navigator-level interactions.

Machine Learning Researcher, Subconscious AI, NY (MLOps | LangChain | GANs | WANDB | GIT | Firestore) (Feb 2024 - Aug 2024)

- Elevated behavioral research through innovation in synthetic data creation, utilizing GANs and VAEs to produce diverse and realistic survey datasets, markedly improving the precision, efficiency, and reliability of experimental findings.
- Spearheaded causal graph modeling initiatives, directly influencing over 15 strategic product decisions, resulting in a projected 20% increase in user engagement through deep insights into customer behaviors and preferences.
- Authored code for a RAG-based Large Language Model to generate real-world product recommendations based on market study simulations run on the product, employing Perplexity API and Azure OpenAI for scalability.

NLP Research Intern, UST Global, India (Hugging Face | Sentiment Analysis | Flask | BERT | Plotly) (Jun 2021 - Aug 2021)

- Conducted research and analysis into state-of-the-art NLP architectures. Designed and executed a sentiment analysis and keyword extraction project, utilizing advanced NLP techniques to classify and extract key features from a review dataset.
- Developed a custom model using the pre-trained DistilBERT architecture, achieving an impressive accuracy of 98% for classification tasks.
- Leveraged the NLTK library and Plotly visualization tool to extract and visualize data, creating a rich and interactive web page using HTML, CSS, and Flask.

PROJECTS

QA ChatBot (NLP | Vector Databases | Databricks Environment | Gen AI | RAG) (Mar 2024 - Jun 2024)

- Led the design and deployment of a QA chatbot utilizing Retriever-Augmented Generation (RAG) and OpenAI models within Databricks, leveraging vector databases for enhanced question-answer processing speed and accuracy.
- Engineered the chatbot for organizational specifics, integrating unique data and knowledge bases, and applied iterative enhancements via NLP techniques to ensure answer precision and relevance.
- Working on construction of a versatile deployment framework to facilitate smooth integration and consistent performance in varied cloud environments.

BERT-Based Patent Analysis (NLP | LLM Finetuning | Tensorflow | Python | Google BigQuery) (Feb 2023 - Mar 2023)

- Applied BERT as a multi-tasking model and fine-tuned it for 3 tasks namely Document Classification, Patent Valuation, and Text Similarity Analysis on Google Patents Public Dataset.
- Attained an F-1 score of 0.75 task-1, calculated score for all the patents which can be used to compare the value of patents in the market for task-2, and computed the Euclidean distance using Faiss Indexing to find the similarity scores of the patents and display the ones most similar to the query for task-3.

Multi-label classification, Multimodal Image Classification and Image Description (Python | Deep Learning | CNN | LSTM) (Apr 2021 - May 2021)

- Synthesized image labels based on their descriptions using a multimodal classification approach by building a combination of CNN and LSTM.
- System was able to recognize patterns and relationships between the images and their descriptions by merging the output of the above networks, resulting in the creation of the most appropriate labels for the images with an accuracy of 80%.

LICENSES & CERTIFICATIONS

Generative AI with Large Language Models Coursera

Issued Aug 2023

Introduction to Machine Learning on AWS Coursera

Issued Aug 2023

Introduction to Generative AI Learning Path Google Cloud skills boost

Neural Networks and Deep Learning Coursera

TECHNICAL SKILLS

Programming languages: Python, C, C++, Javascript

Web Development: HTML5, CSS3, BOOTSTRAP, PHP, Node.js, React, MySQL, MongoDB, flask, Django

AI Frameworks & Big Data: Tensorflow, keras, PyTorch, Pandas, Scikit learn, PySpark, Hadoop

Artificial Intelligence : Deep Learning (Neural Networks), Machine Learning (Supervised , Unsupervised Learning Algorithms), NLP